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Notation

$x_1, x_2, \ldots, x_n$ → data points

$z_1, z_2, \ldots, z_n$ → cluster indicator
$z_i \in \{1, \ldots, k\}$

Goodness of cluster assignment

$$f(z_1, \ldots, z_n) = \sum_{i=1}^{n} \| x_i - \mu_{z_i} \|^2$$

$$\mu_k = \frac{\sum_{i=1}^{n} x_i \, \mathbb{I}(z_i = k)}{\sum_{i=1}^{n} \mathbb{I}(z_i = k)}$$

$\mathbb{I}(u) = \begin{cases} 1, & \text{if } u \text{ is true} \\ 0, & \text{if false} \end{cases}$

Goal

$$\min_{\{z_1, \ldots, z_n\}} \sum_{i=1}^{n} \| x_i - \mu_{z_i} \|^2$$

But k^n possibilities

NP-HARD problem

K-means (Lloyd's Algorithm)

Initialization: $z_1^0, z_2^0, \ldots, z_n^0 \in \{1, 2, \ldots, k\}$

Until convergence:

- Compute means:

$$\forall k, \mu_k^t = \frac{\sum_{i=1}^{n} x_i \, \mathbb{1}(z_i = k)}{\sum_{i=1}^{n} \mathbb{1}(z_i = k)}$$

- Reassignment:

$$\forall i, z_i^{t+1} = \operatorname{argmin} \| x_i - \mu_k^t \|^2$$

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Convergence of Lloyd's algorithm

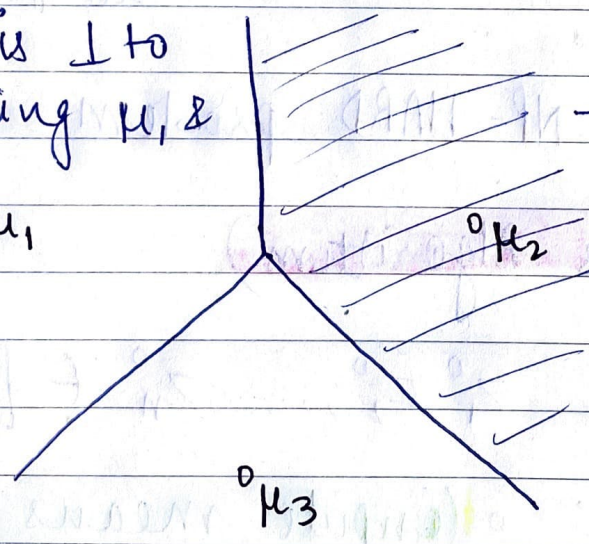
- In every iteration, only those data points change clusters that were unhappy.
- Every reassignment reduces the value of objective function.
- Every reassignment results into a new partition.
- The finiteness of the number of partitions allows us to argue that the algorithm converges.

Nature of clusters

- A half-space is formed by perpendicular bisector of the line joining any two cluster centers.

- Every cluster region is formed by the intersection of k-1 half spaces.

- Boundry is ⊥ to the line joining μ_1 & μ_2 at $\frac{\mu_1 + \mu_2}{2}$



→ Voronoi regions

- If the value of the objective function strictly reduces, it indicates that partitions cannot repeat.

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INITIALISATION OF CLUSTERS

METHOD 1: Uniformly at random

METHOD 2: K-Means ++

Idea: cluster centers should be as far apart as possible

1. Choose μ_1^0 uniformly at random out of $x_1, x_2, \ldots, x_n$.

2. For $l = 2, 3, \ldots, k$,
   Choose μ_i probabilistically proportional to score (S)

$$\forall x \quad S(x) = \min_{j=1,2,\ldots l-1} \|x - \mu_j^0\|^2$$

CHOICE OF K

obj + penalty

Techniques to fix suitable penalty

- Akaike information criterion
- Bayesian information criterion